

Chemistry Lesson Plan: Awesome Everyday Chemistry

- Topic 1: Cool Chemical Reactions
 - https://kids.kiddle.co/Chemical_reaction
 - Fact: A chemical reaction is a process where one or more substances change into new and different substances.
 - Fact: When iron and oxygen mix, they create rust.
 - Fact: Mixing vinegar and baking soda makes carbon dioxide gas, which causes fizzing.
 - Fact: Burning wood in a campfire is a chemical reaction.
 - Fact: When you use or charge batteries, chemical reactions are taking place.
 - <https://littlebinsforlittlehands.com/fizzing-science-experiments-kids-chemistry/>
 - Fact: A chemical reaction happens when two or more substances react to form a new substance with different properties.
 - Fact: A chemical change is usually irreversible.
 - Fact: Gas being produced (bubbles or fizzing) can signal a chemical reaction.
 - Fact: A color change can signal a chemical reaction.
 - Fact: Cooking food, baking bread, and rust forming are everyday chemical reactions.
 - <https://www.sciencebuddies.org/blog/teach-chemical-reactions>
 - Fact: Elephant Toothpaste creates a fun, high-impact foaming demonstration.
 - Fact: The Elephant Toothpaste reaction mixes hydrogen peroxide with yeast and dish soap.
 - Fact: There is no actual toothpaste involved in the Elephant Toothpaste reaction.
 - Fact: Chemical reactions can be explored with simple hands-on classroom experiments.
 - Fact: Reactions can release energy, sometimes producing foam, heat, or gas.
 - <https://www.polygence.org/blog/chemistry-experiments-for-kids>
 - Fact: Magic Milk uses milk, food coloring, and dish soap to create swirling colors.
 - Fact: A baking soda and vinegar volcano erupts thanks to a chemical reaction.

- Fact: Invisible ink can be made with lemon juice and revealed with heat.
 - Fact: A rainbow in a jar works by layering liquids of different densities.
 - Fact: Homemade slime forms when glue mixes with a borax solution.
- Topic 2: States of Matter
 - https://kids.kiddle.co/State_of_matter
 - Fact: The three main states of matter are solid, liquid, and gas.
 - Fact: In a solid, particles are packed tight and stay in fixed spots, just vibrating.
 - Fact: In a liquid, particles are still close but can slide past each other.
 - Fact: In a gas, particles are far apart and move around freely.
 - Fact: How the particles are arranged determines the state of matter.
 - <https://www.theteachingastrophysicist.com/post/7-amazing-states-of-matter-facts>
 - Fact: Water can exist as a solid, liquid, and gas at the same time.
 - Fact: Dry ice does not melt into a liquid; it skips straight to gas.
 - Fact: The change from solid straight to gas is called sublimation.
 - Fact: Temperature and pressure both affect the state of a substance.
 - Fact: Everyday weather, like rain and snow, shows matter changing states.
 - <https://bereanbuilders.com/ecommerce/states-of-matter-homeschool-learning-resources/>
 - Fact: How atoms and particles are arranged in a substance determines its state.
 - Fact: The three primary states of matter are solid, liquid, and gas.
 - Fact: Solids keep a fixed shape because their particles hardly move.
 - Fact: Liquids take the shape of their container but keep the same volume.
 - Fact: Gases spread out to fill whatever space they are in.
 - <https://www.brightinthemiddle.com/lesson-plans-on-states-of-matter/>
 - Fact: By middle school, most students have heard about solids, liquids, gases, and maybe plasma.
 - Fact: Plasma is a fourth state of matter found in stars and lightning.

- Fact: Hands-on demonstrations make states of matter easy to understand.
 - Fact: Adding or removing energy causes matter to change state.
 - Fact: Ice, water, and steam are all the same substance in different states.
- Topic 3: The Periodic Table and Elements
- https://kids.kiddle.co/Periodic_table
 - Fact: Only elements up to atomic number 94 exist naturally on Earth.
 - Fact: The remaining elements are made by scientists in laboratories.
 - Fact: The periodic table organizes all the known chemical elements.
 - Fact: Elements are arranged in rows called periods and columns called groups.
 - Fact: Each element has its own symbol, like O for oxygen and H for hydrogen.
- <https://www.goodfellow.com/eu/resources/the-periodic-table-of-elements-for-kids/>
 - Fact: Earth's core is made mostly of nickel and iron.
 - Fact: Element 101 is named after Mendeleev; it's called mendelevium.
 - Fact: Whoever discovers a new element usually gets to name it.
 - Fact: Elements have been named after people, places, and mythology.
 - Fact: Mendeleev is often called the father of the periodic table.
- <https://wardsworld.wardsci.com/chemistry/chemistry-elements-fun-facts-and-free-poster>
 - Fact: Hydrogen is the most common element in the universe.
 - Fact: Lithium is the lightest of all the metals and can float on water.
 - Fact: Oxygen is the third most abundant element in the universe.
 - Fact: Sodium is never naturally found in its pure form on Earth.

- Fact: Neon is the fifth most common element in the universe.
- https://www.ducksters.com/science/periodic_table.php
 - Fact: There are seven (or eight) total periods on the periodic table.
 - Fact: The first period has only two elements, hydrogen and helium.
 - Fact: The sixth period has 32 elements.
 - Fact: Carbon can form up to 10 million different compounds.
- Fact: Francium is the rarest element on Earth.

All Facts (Color-Coded by Topic)

Topic 1: Chemistry Experiments for Kids

Magic Milk uses milk, food coloring, and dish soap to create swirling colors.

A baking soda and vinegar volcano erupts thanks to a chemical reaction.

Invisible ink can be made with lemon juice and revealed with heat.

A rainbow in a jar works by layering liquids of different densities.

Homemade slime forms when glue mixes with a borax solution.

Topic 2: States of Matter

The three main states of matter are solid, liquid, and gas.

In a solid, particles are packed tight and stay in fixed spots, just vibrating.

In a liquid, particles are still close but can slide past each other.

In a gas, particles are far apart and move around freely.

Plasma is a fourth state of matter found in stars and lightning.

Dry ice does not melt into a liquid; it skips straight to gas.

The change from solid straight to gas is called sublimation.

Adding or removing energy causes matter to change state.

Ice, water, and steam are all the same substance in different states.

Temperature and pressure both affect the state of a substance.

Topic 3: The Periodic Table and Elements

Only elements up to atomic number 94 exist naturally on Earth.

The remaining elements are made by scientists in laboratories.

Elements are arranged in rows called periods and columns called groups.

Each element has its own symbol, like O for oxygen and H for hydrogen.

Mendeleev is often called the father of the periodic table.

Element 101 is named after Mendeleev; it's called mendelevium.

Hydrogen is the most common element in the universe.

Lithium is the lightest of all metals and can float on water.

Carbon can form up to 10 million different compounds.

Francium is the rarest element on Earth.

Lesson Summary

This lesson introduces middle school students to chemistry through three fun, hands-on topics. First, students explore exciting chemistry experiments for kids, such as magic milk, baking soda and vinegar volcanoes, invisible ink, a rainbow in a jar, and homemade slime, which show that chemical reactions and physical properties can be both surprising and easy to observe. Next, students learn about the states of matter, discovering that solids, liquids, and gases differ because of how their particles are arranged, that plasma is a fourth state found in stars and lightning, and that adding or removing energy through changes in temperature and pressure causes substances like ice, water, and steam to change state, sometimes skipping straight from solid to gas through sublimation. Finally, students investigate the periodic table and elements, learning that elements are organized into periods and groups, that each has its own symbol, that only elements up to atomic number 94 occur naturally while the rest are made in laboratories, and that Mendeleev is celebrated as the father of the periodic table. Fun facts, such as hydrogen being the most common element in the universe, lithium being light enough to float on water, carbon forming millions of compounds, and francium being extremely rare, help make chemistry memorable and relatable. Together, these topics give students a playful yet solid foundation in how matter behaves and how the building blocks of the universe are organized.

Topic Images

